

Jordan Carvalho

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EDUCATION

Western University – Ivey Business School (GPA 3.9)

London, ON

Software Engineering & Honours Business Administration Candidate, Dual Degree Expected Graduation: April 2028

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Design, Software Construction, Calculus 3, Linear Algebra, Probability & Statistics
- Self-Directed Study: MIT OpenCourseWare – Topics in Mathematics with Applications in Finance
- **Activities:** Men's Varsity Curling Team (First String) and Western Pickleball Club
- **Awards:** OUA Academic Achievement Award, U SPORTS Academic All-Canadian (2025), Dean's Honour List (2025, 2026), Scholarship of Distinction (\$3,500), Ontario Winter Games (Mixed Doubles)–Silver Medal

PROFESSIONAL AND LEADERSHIP EXPERIENCE

Aquatic Supervisor

Thornhill, ON

Bayview Golf and Country Club

May 2022 – Present

- **Earned two promotions** over five summers—from Lifeguard & Swim Instructor to Deck Supervisor to Aquatic Supervisor—through consistent operational performance and demonstrated leadership.
- Led daily operations of the aquatic facility by **supervising a team of 12 lifeguards**, enforcing safety protocols, and coordinating emergency preparedness, maintaining high safety and service standards throughout the season.
- Conducted interviews, hired **3 of 5 candidates**, built weekly staff schedules, and trained **12 lifeguards**, building a well-prepared team capable of delivering consistent service and maintaining operational coverage.
- Planned and coordinated aquatic programming and major club events, collaborating with multiple departments to deliver engaging experiences for over **100 members** during peak events such as Canada Day.
- Resolved staff conflicts and member concerns through proactive communication and problem-solving, fostering a collaborative work environment while maintaining high levels of member service and operational efficiency.

PROJECTS

AI Trading Strategy

Independent Project (Python, XGBoost, pandas, quantitative finance)

- Built end-to-end machine learning Python and XGBoost to predict 5-day directional stock movements using engineered features including momentum, volatility breakout, and mean reversion signals.
- Implemented walk-forward backtesting framework with rolling model retraining, transaction cost modeling, and position-holding constraints to prevent lookahead bias and simulate realistic live-trading conditions.
- Achieved 0.77 Sharpe ratio with -12.4% maximum drawdown, materially outperforming the market benchmark's -38.5% drawdown over the same test period.
- Designed the full pipeline end-to-end: data ingestion, feature engineering, model training, evaluation, and performance visualization.

AI Financial Sentiment Analysis Platform

Independent Project

- **Engineered** Python backend systems to process 500+ Reddit and Twitter posts, developing scalable data pipelines for financial sentiment analysis and preprocessing workflows.
- Integrated Ollama-based NLP models to summarize financial discussions and generate community sentiment scores while filtering irrelevant content.
- Optimized SQL database queries through restructuring and database tuning improving backend data retrieval.
- Evaluated multiple AI models through iterative experimentation to identify the highest performing approach for financial sentiment analysis.

iSky Airline Management System

Software Design (Java, JavaFX, Vaadin, Spring Data JPA, H2)

- Designed and implemented a full-stack airline management platform end-to-end, modeling system architecture through UML use case, class, and sequence diagrams to formalize functional requirements and object interactions.
- Built a role-based access control (RBAC) system integrated with a persistent H2 database view Spring Data JPA, enabling secure user management and dynamic UI behavior based on user roles.
- Developed a JavaFX desktop application for business manager operations (flight management, secure authentication) and a Vaadin web application for passenger reservation workflows.

SKILLS & INTERESTS

Languages: Python, Java, JavaScript, SQL, HTML5, CSS3, C# | **Frameworks:** React, Node.js, Next.js, Unity **Data & AI:** NLP, Machine Learning Algorithms, Web Scraping, Ollama, JSON Parsing | **Tools:** Visual Studio Code, Supabase, IntelliJ IDEA, Git, GitHub, Scene Builder, Excel | **Interests:** Pickleball, Golf, Tennis, Swimming, Fitness, Formula 1